Medical Physics

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Chapter 10:
Special Relativity

Chapter 10： Special Relativity

You will learn:

1. The two postulates of Einstein's special theory of relativity, and what motivation these postulates.
2. Why different observers can disagree about whether two events are simultaneous.
3. How relativity predicts that moving clocks run slow, and experimental evidence that confirms this.
4. How the length of an object changes due to the object’s motion.
5. How the velocity of an object depends on the frame of reference from which it is observed.
6. How the theory of relativity modifies the relationship between velocity and momentum.
7. How to solve problems involving work and kinetic energy for particles moving at relativistic speeds.
8. Some of the key concepts of Einstein’s general theory of relativity.

Chapter 10： Special Relativity

Galilean transformation of coordinate

$$\begin{cases} x' = x - vt \\ y' = y \\ z' = z \\ t' = t \end{cases}$$

Time dilation

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - v^2/c^2}}$$

Length contraction

$$L = L_p\sqrt{1 - v^2/c^2}$$

Linear moment

$$p = \frac{mv}{\sqrt{1 - v^2/c^2}}$$

Kinetic energy

$$K = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2$$

Total energy

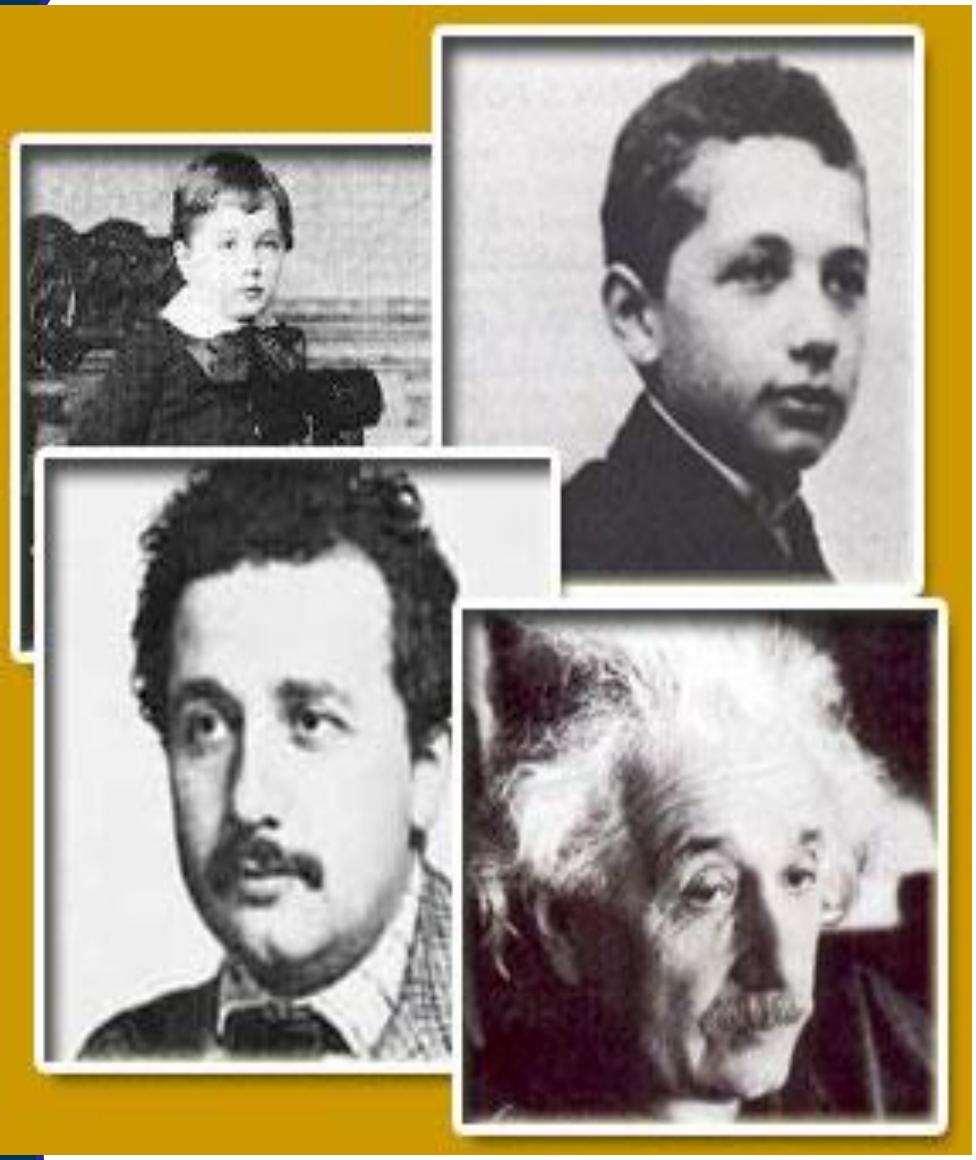
$$E = K + mc^2 = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

$$E^2 = (pc)^2 + (mc^2)^2$$

The relativistic mass

$$\frac{m}{\sqrt{1 - v^2/c^2}}$$

Chapter 10： Special Relativity



Einstein（1879-1955）

A German-born theoretical physicist, who developed the theory of relativity, one of the two pillars of modern physics (alongside quantum mechanics. He is best known to the general public for his mass–energy equivalence formula, which has been dubbed "the world's most famous equation". He received the 1921 Nobel Prize in Physics "for his services to theoretical physics, and especially for his discovery of the law of the photoelectric effect”.

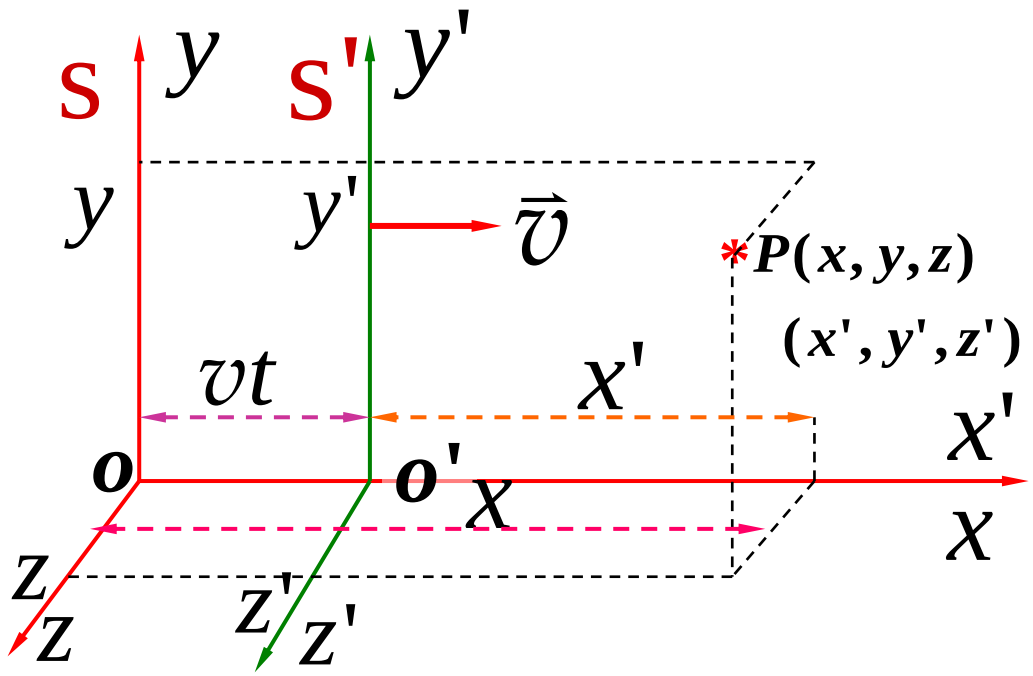
https://en.wikipedia.org/wiki/Albert_Einstein

10.1 the principle of Newtonian relativity

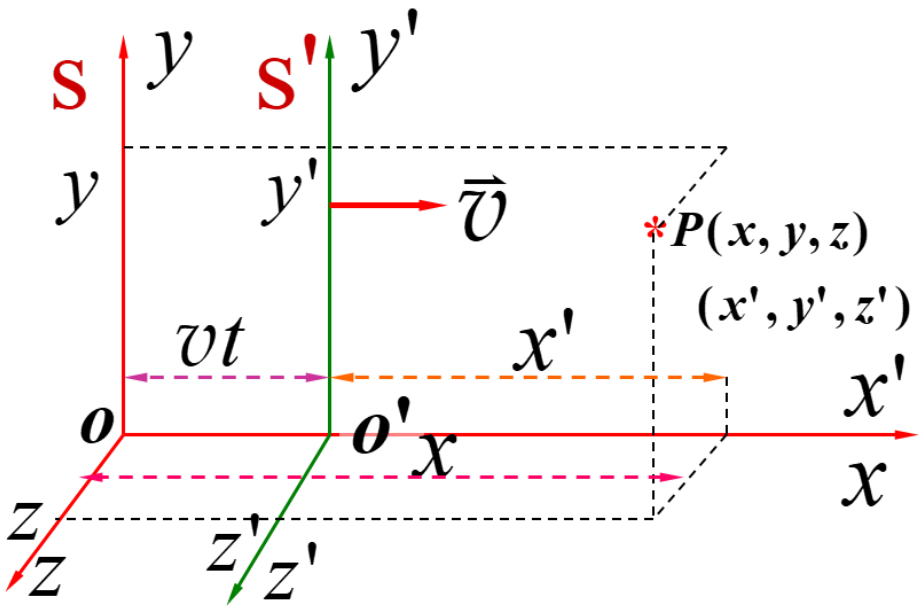
The Relativity is about the relationship among the time, space and material introduced by Einstein.

The principle of Newtonian relativity states that:

The laws of mechanics must be the same in all inertial frames of reference.



10.1 the principle of Newtonian relativity

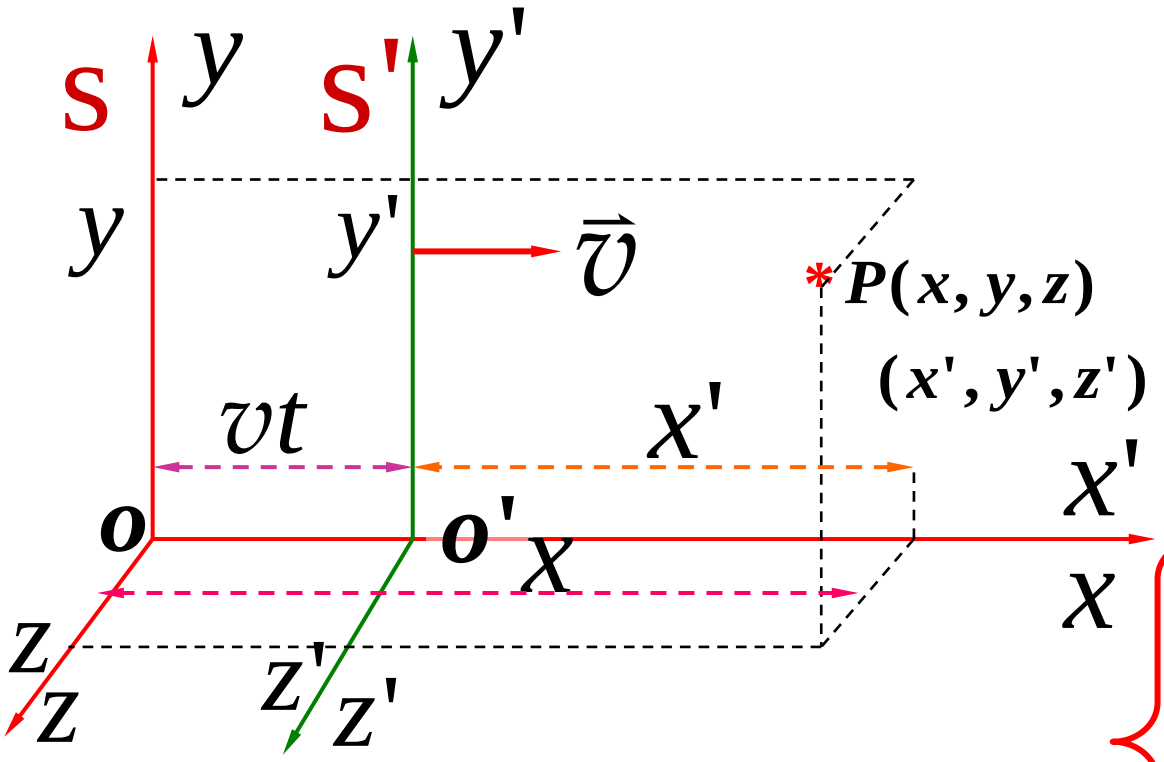


An observer in S describes an event with space-time coordinates (x, y, z, t), while an observer in *S'* uses (x', y', z', t') to describe the same event. These coordinates are related by the equations called Galilean transformation of coordinate.

$$\begin{cases} x' = x - vt \\ y' = y \\ z' = z \\ t' = t \end{cases} \quad \textbf{or} \quad \begin{cases} x' = x + vt \\ y' = y \\ z' = z \\ t' = t \end{cases}$$

10.1 the principle of Newtonian relativity

Galilean velocity transformation:

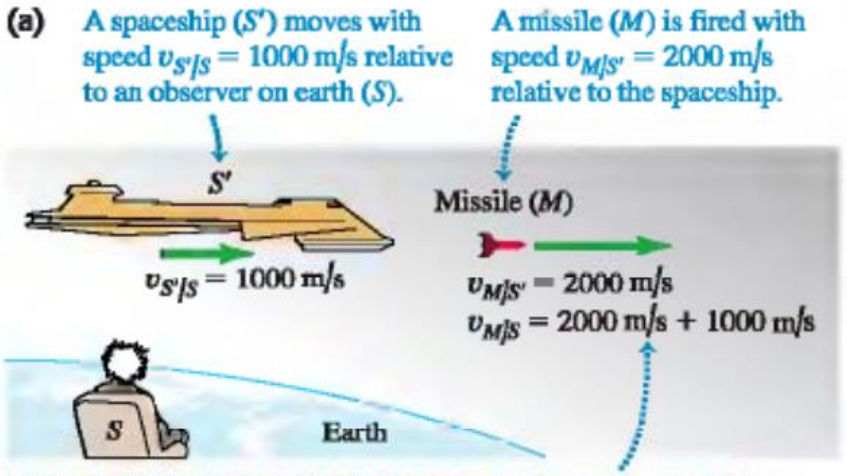


$$u'_x = u_x - v$$

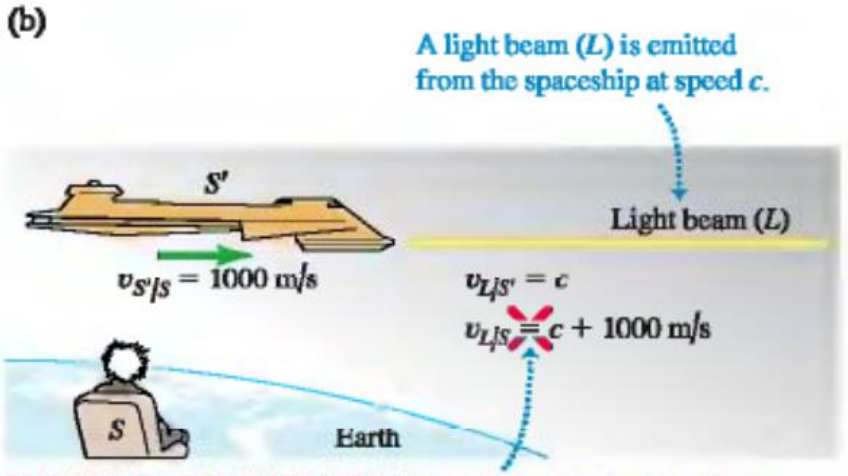
$$u'_y = u_y$$

$$u'_z = u_z$$

10.1 the principle of Newtonian relativity



NEWTONIAN MECHANICS HOLDS: Newtonian mechanics tells us correctly that the missile moves with speed $v_{M|S} = 3000$ m/s relative to the observer on earth.

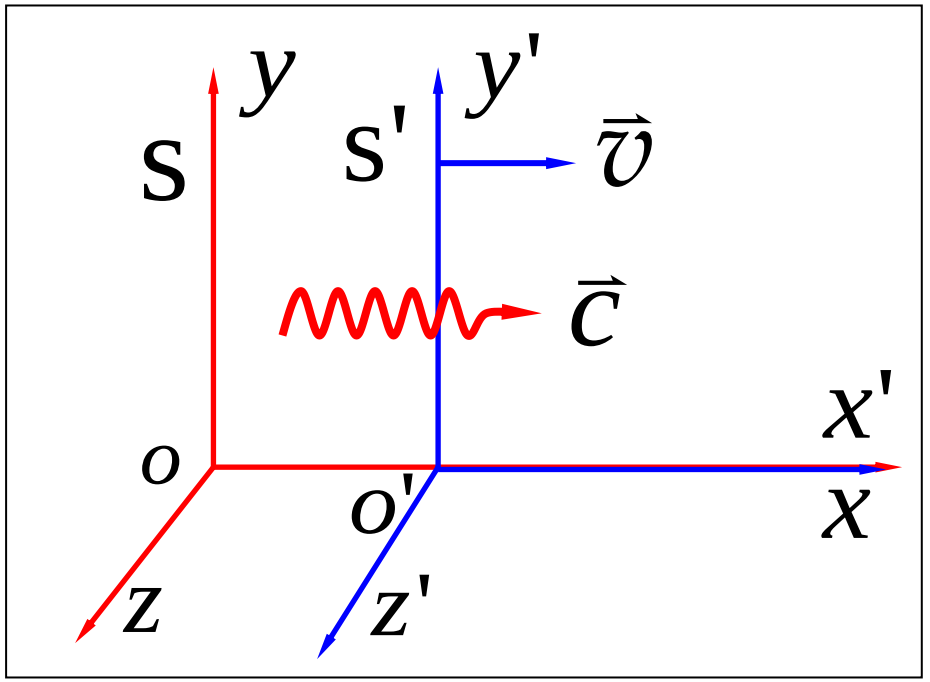


NEWTONIAN MECHANICS FAILS: Newtonian mechanics tells us *incorrectly* that the light moves at a speed greater than c relative to the observer on earth ... which would contradict Einstein's second postulate.

10.1 the principle of Newtonian relativity

$$\vec{c}' = \vec{c} \pm \vec{v}?$$

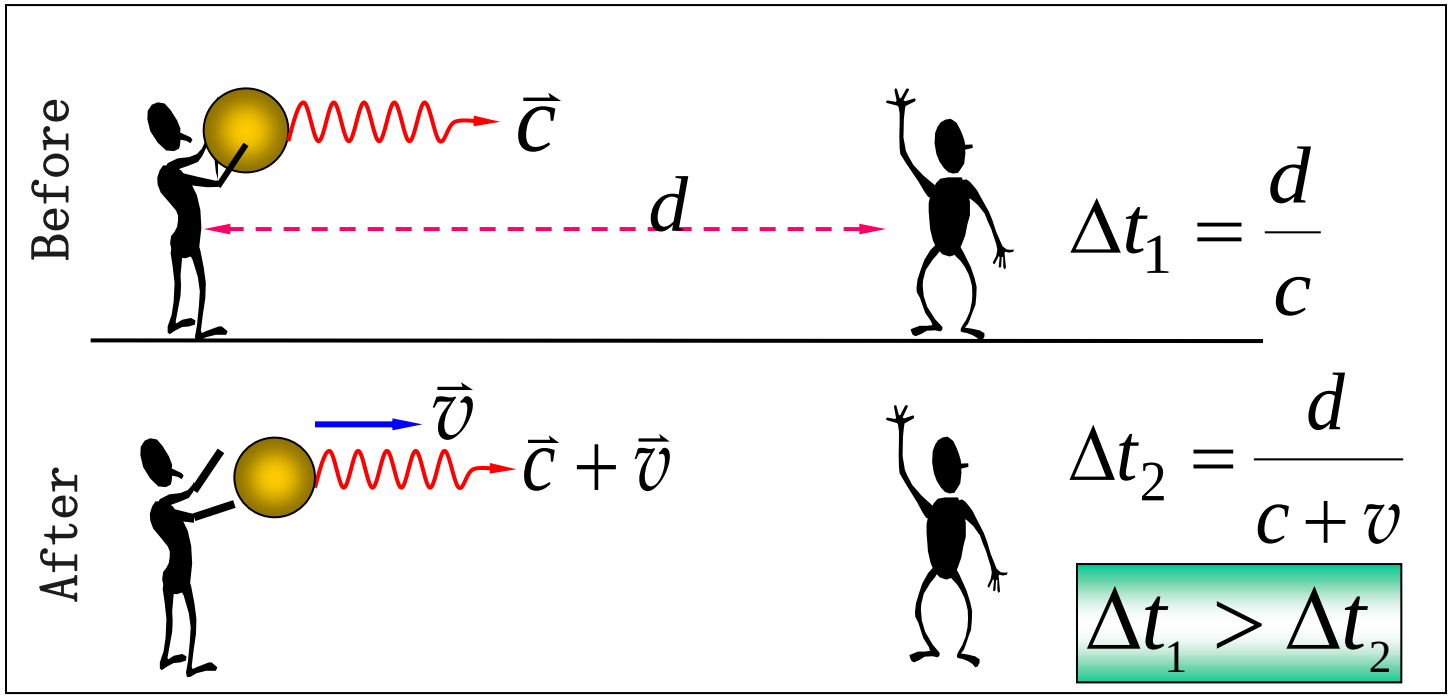
$$C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$



C=2.998 × 10^8 m s^{-1}

10.1 A Paradox

(Galilean transformation)



Result: observer see the ball was throw out first, then see the ball was throw! This is a paradox!

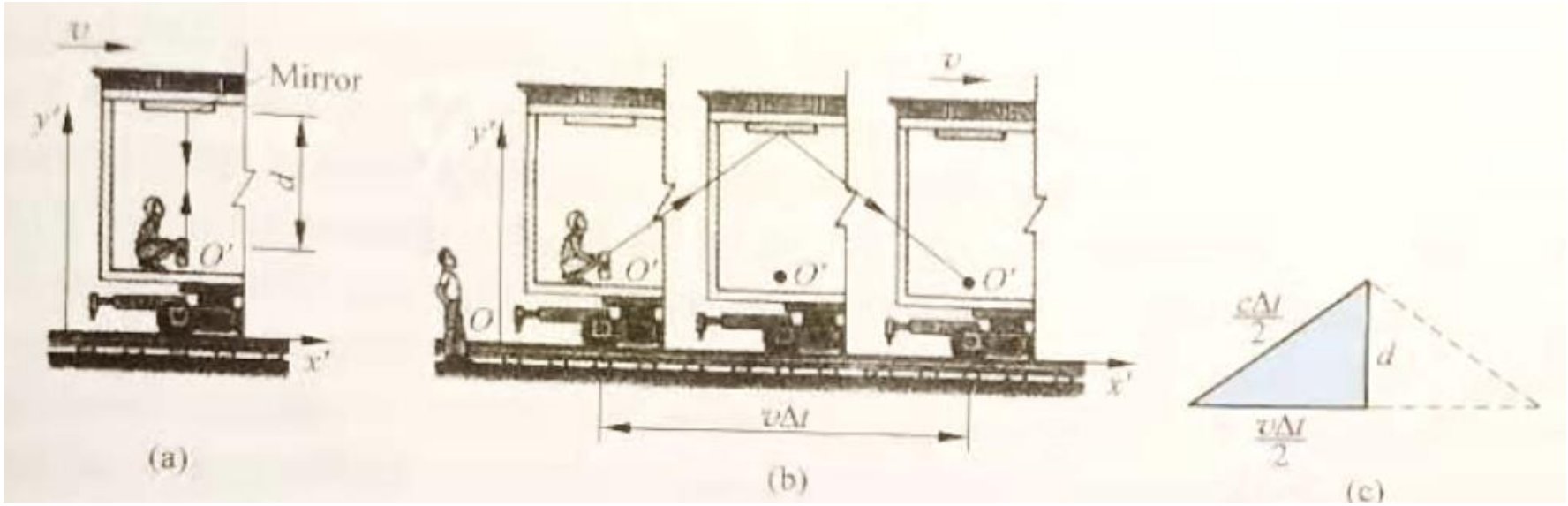
10.1.2 Einstein 2 postulates of relativity

Einstein's special theory of relativity is based on two postulates,

1. The principle of relativity: All laws of physics are the same for observers in all inertial reference. No one frame is preferred over any other.

2. The constancy of the speed of light: the speed of light in vacuum (free space)has the same value c ($c=3.0*10^8$ m s^{-1}), in all inertial frames, regardless of the velocity of the observer or the velocity of the source emitting the light.

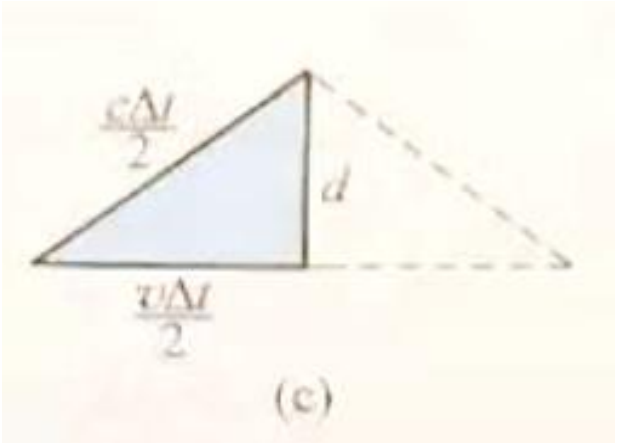
10.2.2 Time dilation



(a) A mirror is fixed to a moving vehicle, and a light pulse leaves O' at rest in the vehicle; (b) Relative to a stationary observer on Earth, the mirror and O' move with a speed v.

Note that the distance the pulse travels is greater than 2d as measured by the stationary observer; (c) The right triangle for calculating the relationship between Δt and $\Delta t'$.

10.2.2 Time dilation



$$\left(\frac{c\Delta t}{2}\right)^2 = \left(\frac{v\Delta t}{2}\right)^2 + d^2$$

$$\Delta t = \frac{2d}{\sqrt{c^2 - v^2}} = \frac{2d}{\sqrt{1 - v^2/c^2}}$$

$$\Delta t' = 2d/c$$

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - v^2/c^2}} = r\Delta t'$$

$$r = \frac{1}{\sqrt{1 - v^2/c^2}}$$

Time dilation

$$\Delta t > \Delta t'$$

10.2.2 Time dilation

(1) Time dilation

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - v^2/c^2}}$$

Therefore. Δt> Δt'. This effect is known as time dilation.

All physical, chemical and biological processes slow down relative to a stationary clock when those processes occur in a moving frame.

10.2.2 Length contraction

(2) Length contraction

The length L_p, of an object measured by an observer in an inertial reference frame in which the object is at rest is called its proper length Observers in frames moving relative to that frame and parallel to that length will measure a shorter length. For an observer moving with relative speed *v*. The measured length is

$$L = L_p\sqrt{1 - v^2/c^2}$$

Therefore, $L<L_p$

This result may be interpreted as follows: $L = L_p\sqrt{1 - v^2/c^2}$

If object has a length *Lp* when it is at rest, then when it moves with speed *v* in a direction parallel to its length, it contracts to the length L, where

10.2.2 Example

Example 1. A spaceship is measured to be 120m long while at rest relative to an observer, If this spaceship now flies by the observer with a speed *0.99c,* what length does the observer measure? If the spaceship moves past the observer with a speed of 0.010c, what length will the observer measure?

10.2.2 Example

Solution: If the spaceship moves with a speed of 0.99c, the length measured by the observer is

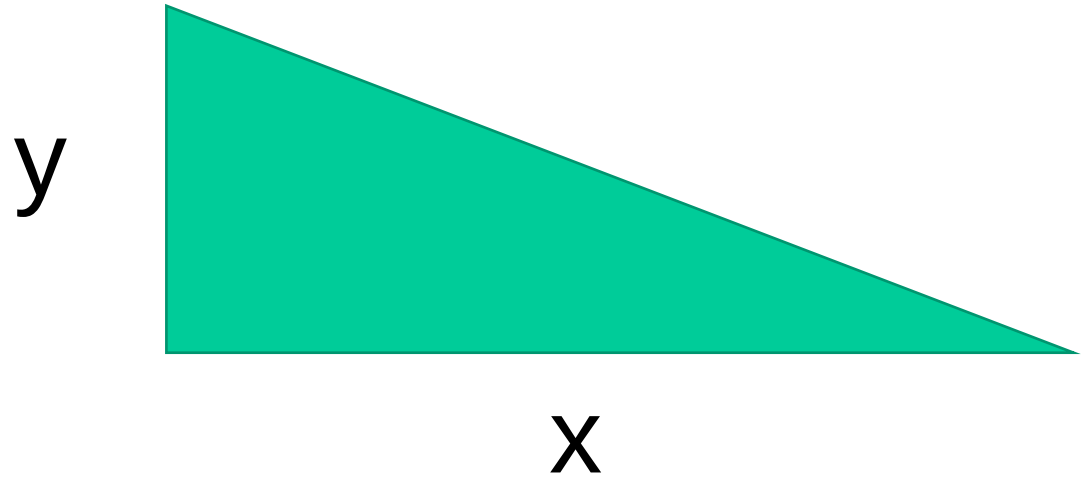
$$L = L_p\sqrt{1 - v^2/c^2} = \mathbf{120\,m}\sqrt{1 - \frac{(0.99c)^2}{c^2}} = \mathbf{17\ m}$$

If the spaceship moves with a speed of 0.010c, the length is

$$L = L_p\sqrt{1 - v^2/c^2} = \mathbf{120\,m}\sqrt{1 - \frac{(0.01c)^2}{c^2}} = \mathbf{119.994\ m}$$

10.2.2 Example 2

A spaceship in the form of a triangle flies by an observer with a speed of 0.95c. When the spaceship is at rest relative to the observer, the distances x and y are measured to be 52m and 25m, respectively. What is the shape of the spaceship as seen by an observer at rest when the spaceship is in motion along the direction shown here.

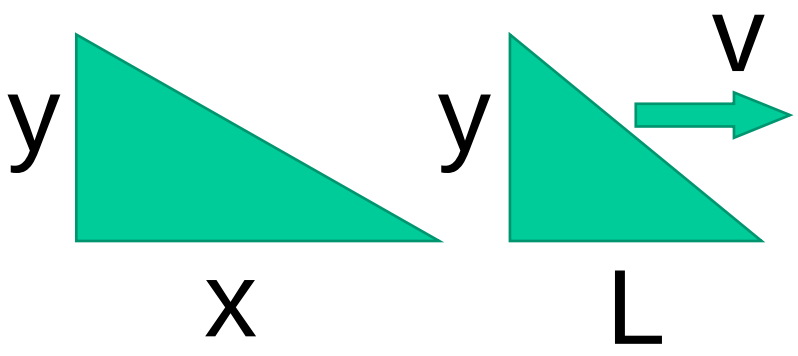


10.2.2 Example 2

Solution: The observer sees the horizontal length of the spaceship to be contracted to a length

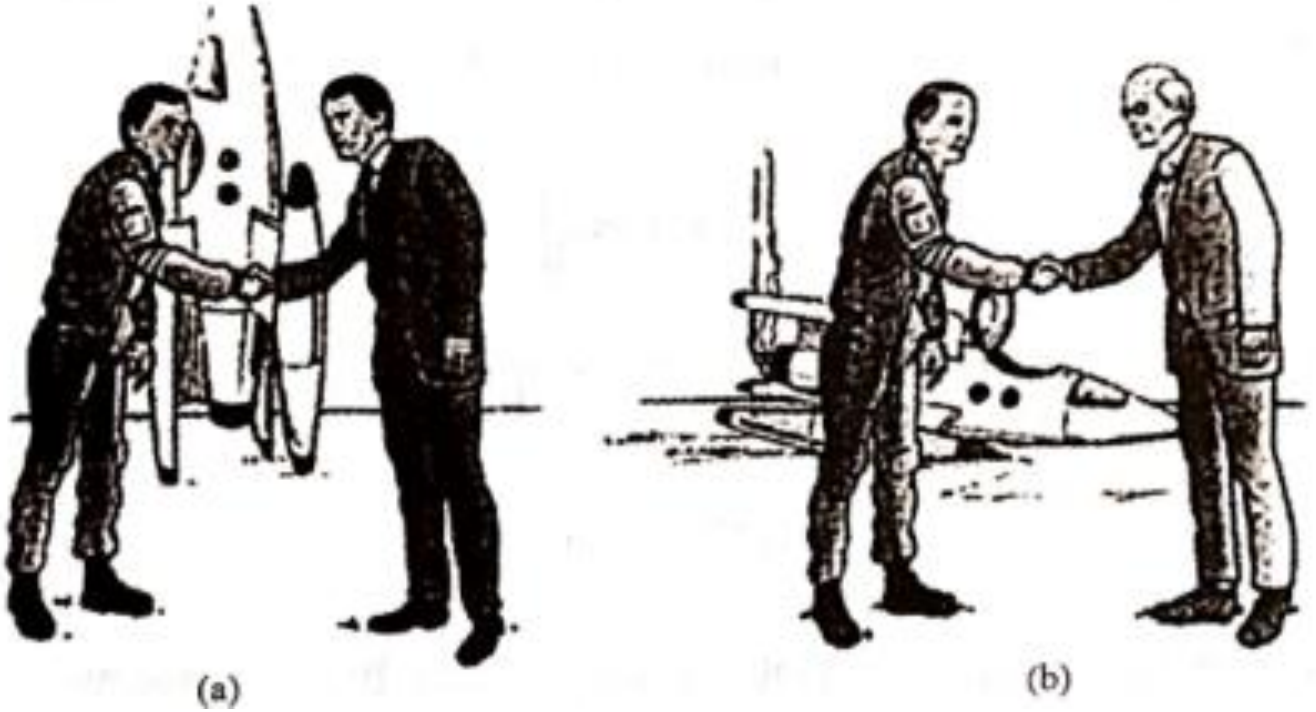
$$L = L_p\sqrt{1 - v^2/c^2} = \mathbf{52m}\sqrt{1 - \frac{(0.95c)^2}{c^2}} = \mathbf{16\ m}$$

the 25m vertical length is unchanged because it is perpendicular the size of the spaceship as seen motion between observer and spaceship.



(a) When the spaceship at rest;
(b) The spaceship appears to look like this when it moves to the right with a speed *v*. Note that only its *x* dimension is contracted in this case.

10.2.2 The twin paradox



As the twins depart, they are the same age; (b) When Speedo returns from his journey to Planet X, he is younger than his twin Goslo, who remained on Earth.

10.3 Momentum and energy

A particle of mass m are valid at any physically possible speed:

Linear moment $$p = \frac{mv}{\sqrt{1 - v^2/c^2}}$$

$$F = \frac{dp}{dt} = \frac{d}{dt}\frac{mv}{\sqrt{1-v^2/c^2}} \qquad \mathbf{a} = \frac{dv}{dt} = \frac{F}{m}\left(1 - \frac{v^2}{c^2}\right)^{3/2}$$

As a particle's speed increases, the acceleration for a given force continuously decreases. As the speed approaches *c*, the acceleration approaches zero, no matter how great the force. Thus it is impossible to accelerate a particle from a state of rest to a speed equal to or greater than *c*, and the speed of light is sometimes referred to as the ultimate speed.

10.3 Momentum and energy

A particle of mass m are valid at any physically possible speed:

Linear moment $$p = \frac{mv}{\sqrt{1 - v^2/c^2}}$$

Kinetic energy $$K = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2$$

Total energy $$E = K + mc^2 = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

Mass energy or Rest energy: mc^2

10.3 Momentum and energy

Mass energy or Rest energy: mc^2

$$E^2 = (pc)^2 + (mc^2)^2$$

For particle at rest, p=0, $E=mc^2$

when no rest mass: m = 0

$$E = pc$$

Massless particles, such as photons, travel with speed of c.

10.3 Momentum and energy

In special relativity, the principle of conservation of energy states as follows：

$$E = \Sigma Ei = \Sigma \left[\frac{m_i c^2}{\sqrt{1 - vi^2/c^2}} \right] = \Sigma m_i c^2 + \Sigma K_i$$

For an isolated system of particles. the total energy E of the system remains constant, no matter what interactions may occur among the particles.

10.3 Momentum and energy

The relativistic mass

$$\frac{m}{\sqrt{1-v^2/c^2}}$$

The conservation of total energy is equivalent to the conservation of relativistic mass. Therefore, although rest mass may not be conserved for an isolated system, relativistic mass is always conserved.

10.3 Example

Example 1. An electron has a kinetic energy *K* of 2.53MeV.
(1) What is its total energy E?
(2) What is its momentum p?
(3) What is the electron's relativistic mass?

Solution: (a) according to

$1\ \text{eV} = 1.6*10^{-19}\ \text{J}$

$$E = K + mc^2$$

$mc^2 = (9.1 * 10^{-31}\ \text{kg})(3.0*10^{8}\ \text{m s}^{-1})^2 = 8.2*10^{-14}\ \text{J}$

$= 5.1 * 10^{5}\ \text{eV} = 0.51\ \text{MeV}$

$E = K + mc^2 = 2.53\ \text{MeV} + 0.51\ \text{MeV} = 3.04\ \text{MeV}$

10.3 Example

(b) According to:

$$E^2 = (pc)^2 + (mc^2)^2$$

$$(pc)^2 = E^2 - (mc^2)^2$$

$$(pc)^2 = (3.04MeV)2 - (0.51MeV)^2$$

$$pc = 3.0\ MeV$$

$$p = 3.0\ MeV/c$$

10.3 Example

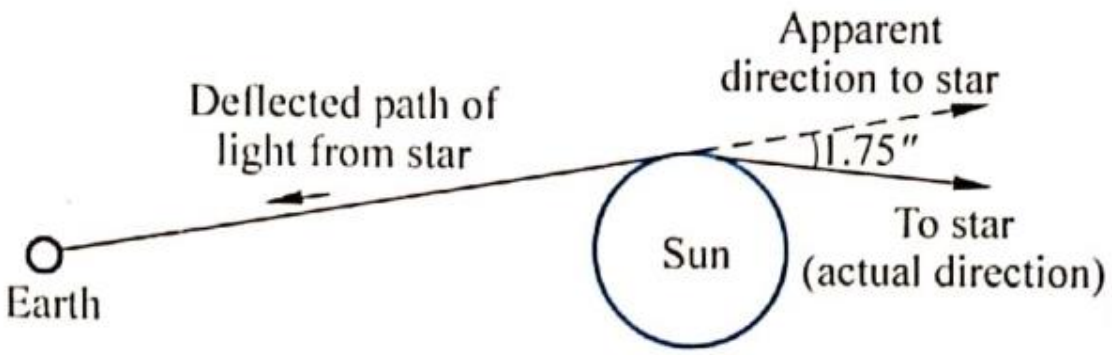
(c) According to:

$$E = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

$$\frac{m}{\sqrt{1-v^2/c^2}} = \frac{E}{C^2} = \frac{E}{mC^2}\, m = \frac{3.04\ MeV}{0.51\ MeV} = 5.96\ m$$

Thus, the electron's relativistic mass is 5.96 times its rest mass

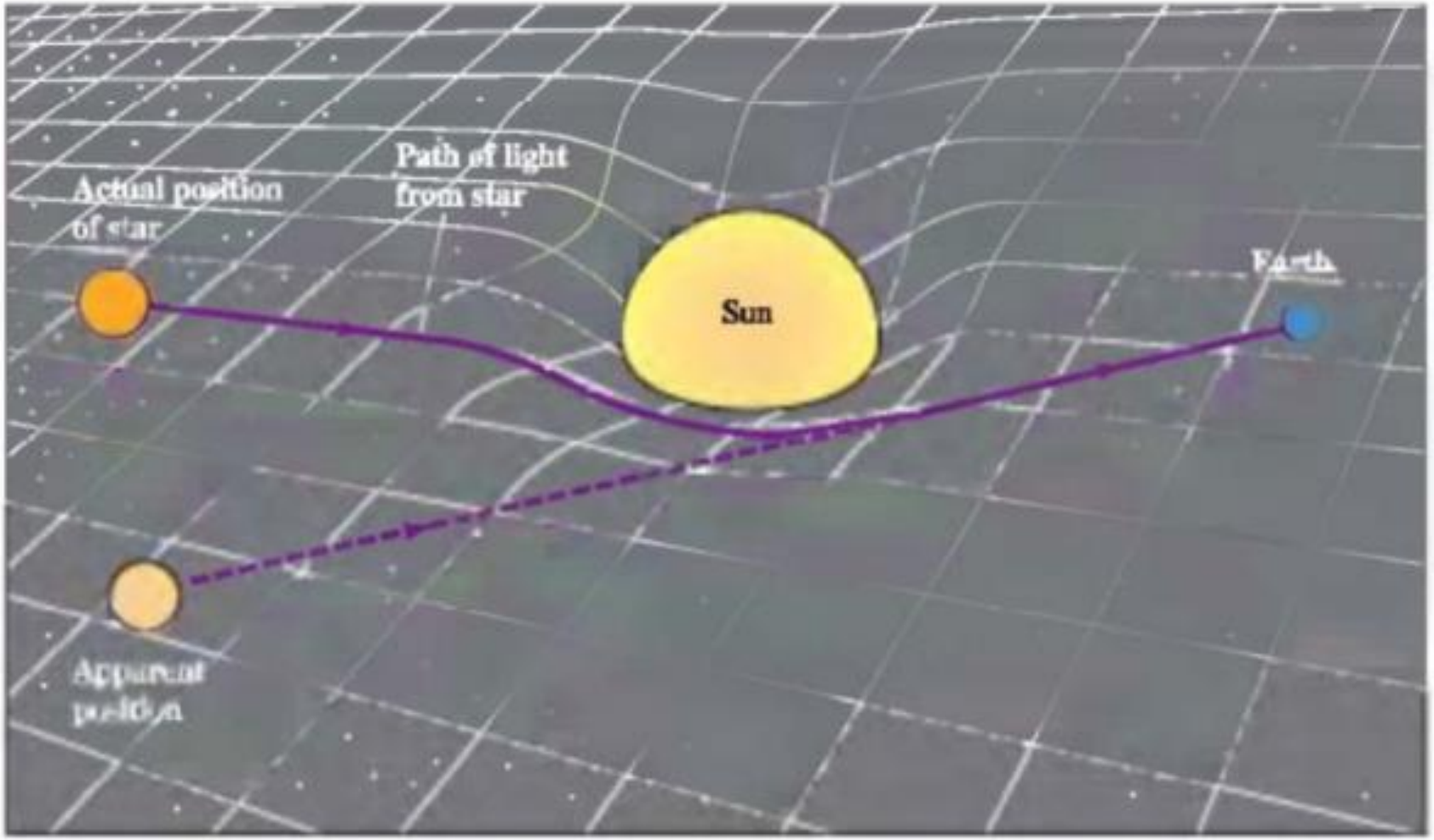
10.3.3 Beyond the Special Relativity



Deflection of starlight passing near the Sun. the Sun and other remote objects can act as a gravitational lens.

The presence of a mass causes a curvature of space-time in the vicinity of the mass, and this curvature dictates the space-time path that all freely moving objects must follow. As one physicist says: "Mass tells space-time how to curve; curved space-time tells mass how to move." One important test of general relativity is the prediction that a light ray passing near the Sun should be deflected by some angle. This prediction was confirmed by astronomers as bending of starlight during a total solar eclipse shortly following World War I.

10.3.3 Beyond the Special Relativity



Questions

1. Two identical clocks are synchronized. One is put in orbit directed eastward around the Earth while the other remains on Earth. Which clock runs slower? When the moving clock returns to Earth, are the two still synchronized?

2. Some of the distant stars, called quasars, are receding from us at half the speed of light (or greater). What is the speed of the light we receive from these quasars?

3. We have seen that if several observers watch two events, labeled *A* and *B*, one of them may say that event *A* occurred first but another may claim that it was event *B* that did so. What would you say to a friend who asked you which event really did occur first?

Chapter 10：

4. From the results of special relativity, would you say that space, time, and distance are absolute quantities?

5. If zero rest mass particles have a speed *c* in one reference frame, can they be found at rest in any other frame? Can such particles have any speed other than *c?*

6. A particle moves with a speed close to the speed of light. A constant force F is applied for a time *t* in the direction of motion. Would you expect the speed change predicted by relativistic mechanics to be greater or less than that predicted by Newtonian mechanics? Explain.

Chapter 10：

7. How is it possible that photons of light, which has zero rest mass, have momentum?

8. If the total relativistic energy E is conserved in a nuclear reaction, why isn't the relativistic kinetic energy also conserved?

9. Physicists generally ignore rest mass energy when analyzing collisions between particles moving at low velocities. Why?